B. President's Office

time limit per test: 2 seconds
memory limit per test: 64 megabytes

President of Berland has a very vast office-room, where, apart from him, work his subordinates. Each subordinate, as well as President himself, has his own desk of a unique colour. Each desk is rectangular, and its sides are parallel to the office walls. One day President decided to establish an assembly, of which all his deputies will be members. Unfortunately, he does not remember the exact amount of his deputies, but he remembers that the desk of each his deputy is adjacent to his own desk, that is to say, the two desks (President's and each deputy's) have a common side of a positive length.

The office-room plan can be viewed as a matrix with n rows and m columns. Each cell of this matrix is either empty, or contains a part of a desk. An uppercase Latin letter stands for each desk colour. The «period» character («`.`») stands for an empty cell.

Input

The first line contains two separated by a space integer numbers n, m $(1 \le n, m \le 100)$ — the length and the width of the office-room, and c character — the President's desk colour. The following n lines contain m characters each — the office-room description. It is guaranteed that the colour of each desk is unique, and each desk represents a continuous subrectangle of the given matrix. All colours are marked by uppercase Latin letters.

Output

Print the only number — the amount of President's deputies.

Examples

input Copy

```
3 4 R
G.B.
.RR.
TTT.
```

output Copy

```
2
```

input Copy

```
3 3 Z
...
.H.
..Z
```

output Copy

```
0
```

Codeforces Beta Round 6 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
368627273	Mar/29/2026 07:30	Accepted

→ Problem tags

implementation *1100

No tag edit access

→ Contest materials

- Codeforces Beta Round #6
- Tutorial #1 (ru)
- Tutorial #2 (en)

Server time: Mar/29/2026 11:34:21UTC+7 (f1).
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